OMANSHU BHATT

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Portfolio: https://omanshu-bhatt-portfolio.vercel.app/ **Github:** https://github.com/Omanshu09 **Linkedin**: https://www.linkedin.com/in/omanshubhatt/

PROFESSIONAL SUMMARY

Computer Engineering graduate with experience building AI/ML, backend and full-stack applications through projects. Built projects involving Retrieval-Augmented Generation pipelines, information retrieval, real-time collaboration and code intelligence with a focus on solving practical engineering problems and building end-to-end applications.

TECHNICAL SKILLS

Languages: Python, C, Java, JavaScript, SQL
Web & Frameworks: React.js, FastAPI, REST APIs, Node.js, Django, HTML, CSS, TypeScript
Data & Databases: Elasticsearch, PostgreSQL, Redis, Neon
AI / ML: PyTorch, LangChain, Generative AI, RAG, Vector Databases
DevOps & Systems: Docker, Kubernetes, Kafka, CI/CD (Jenkins, GitHub Actions)
Distributed & Realtime Systems: Yjs, CRDTs, Ably, IndexedDB, WebRTC
Testing & QA: Manual Testing, Unit Testing, Integration Testing, API Testing, Postman, Pytest, Jest, SDLC, STLC
Core CS: Data Structures & Algorithms, Operating Systems, Database Management Systems, Object-Oriented Programming

FEATURED PROJECTS

Web Research & Evidence Engine

- Built a web research engine for multi-source search, content extraction, relevance filtering, ranking and evidence-based answer generation using LLMs.
- **Tech:** Python, FastAPI, React, Web Search, Information Retrieval, BM25, RAG, LLMs, Docker
- Benchmarked across 100+ research queries and presented at the Agents for Humans Hackathon | Amazon/AWS

Code: https://github.com/Omanshu09/Search-Engine **Live:** https://search-engine-frontend-xi.vercel.app/

Real-Time Collaborative Engineering Workspace

- Built a real-time collaborative workspace with multi-user CRDT-based editing, near-instant synchronization, independent Python code execution, shared output and offline state recovery.
- **Tech**: React, TypeScript, Vite, Node.js, Express, Yjs, CRDTs, Ably, IndexedDB, PostgreSQL, Neon, E2B, REST APIs, Vercel, Render
- Instant collaborative synchronization during multi user editing, with offline local state retention and isolated Python execution

Code: https://github.com/Omanshu09/Nexus-deployment **Live:** https://nexus-deployment-frontend.vercel.app/

Real-Time Translated Video Calling Platform

- Built a video calling platform that enables participants to speak different languages while seeing and hearing real-time translations, using Whisper STT and NLLB-200 entirely on-device via WebGPU with a WASM fallback.
- **Tech:** React, TypeScript, Vite, WebRTC, Web Workers, Whisper STT, NLLB-200, WebGPU, Web Speech API, Vercel
- Tested multi-user editing, offline recovery and isolated Python execution across sessions to verify synchronization and state consistency.

Code: https://github.com/Omanshu09/Fluo-Call **Live:** https://fluo-call.vercel.app/

EXPERIENCE

Trainee Intern | Indian Oil Corporation Limited (IOCL) | July 2025 – August 2025

- Worked with operational data and database systems to analyze trends and recurring patterns in engineering workflows.
- Collaborated with IOCL engineers and technical teams, gaining experience with database management, enterprise engineering processes and quality-focused problem solving.

Virtual Intern — AI/ML & Robotics | Kodacy | July 2024 – August 2024

- Applied Python to AI/ML and data-analysis tasks, working with datasets to identify patterns, evaluate results and develop practical solutions.
- Developed Python-based robotics simulation prototypes involving algorithmic problem solving and automation.

CERTIFICATIONS

- Oracle — Agentic AI Certified Foundations Associate
- TCS iON Career Edge - Young professional
- NASA — Hyperspectral Data for Land and Coastal Systems
- Cisco — Data Science

Virtual Experience

- Commonwealth Bank — Software Engineering Job Simulation
- University of Oxford — AI, Justice and rule of law
- BCGX — GenAI Job Simulation
- Deloitte — Cyber Job Simulation

EDUCATION

- **Govind Ballabh Pant University of Agriculture and Technology, Pantnagar | 2022 – 2026 | CGPA: 7.2**
  Bachelor of Technology (B.Tech) - Computer Engineering
- **Beersheba Senior Secondary School, Almora**
  Class XII | 2021 – 2022 | 90.4%
  Class X | 2019 – 2020 | 95.3